b) $\forall x(C(x) \wedge F(x))$: For ~~t~~all people, a person is a comedian and is also Funny.

c) $\exists x(C(x) \rightarrow F(x))$: There exists a person from the set of all people, who if is a comedian, then is also funny.

d) $\exists x(C(x) \wedge F(x))$: There exists a person from the set of all people who is a comedian and is also funny.

9) Let $P(x)$ be the statement "x can speak Russian" and let $Q(x)$ be the statement "x Knows the computer language C++." Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers and logical connectives. The domain for quantifiers consists of all students at your school.

a) There is a student at your school who can speak Russian and who knows C++ : $\exists x(P(x) \wedge Q(x))$

b) There is a student at your school who can speak Russian but who doesn't know C++ : $\exists x(P(x) \wedge \neg Q(x))$

c) Every student at your school either can speak Russian or knows C++ : $\forall x(P(x) \vee Q(x))$

d) No student at your school can speak Russian or knows C++ : $\neg \exists x(P(x) \vee Q(x))$

10) Let $C(x)$ be the statement "x has a cat", let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret". Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers and logical connectives. Let the domain consist of all students in your class.

a) A student in your class has a cat, a dog and a ferret: $\exists x(C(x) \wedge D(x) \wedge F(x))$

b) All students in your class have a cat, a dog and a ferret: $\forall x(C(x) \wedge D(x) \wedge F(x))$

c) Some student in your class has a cat and a ferret, but not a dog: $\exists x(C(x) \wedge F(x) \wedge \neg D(x))$